

TEXT0 12- UNIVERSITY PHYSICS WITH MODERN PHYSICS – PART 2

University Physics with Modern Physics

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CHAPTER 1 - UNITS, PHYSICAL QUANTITIES, AND VECTORS

The study of physics is important because physics is one of the most fundamental of the sciences. Scientists of all disciplines make use of the ideas of physics, including chemists who study the structure of molecules, paleontologists who try to reconstruct how dinosaurs walked, and climatologists who study how human activities affect the atmosphere and oceans. Physics is also the foundation of all engineering and technology. No engineer could design a flat-screen TV, an interplanetary spacecraft, or even a better mousetrap without first understanding the basic laws of physics.

The study of physics is also an adventure. You will find it challenging, sometimes frustrating, occasionally painful, and often richly rewarding and satisfying. It will appeal to your sense of beauty as well as to your rational intelligence. If you've ever wondered why the sky is blue, how radio waves can travel through empty space, or how a satellite stays in orbit, you can find the answers by using fundamental physics. Above all, you will come to see physics as a towering achievement of the human intellect in its quest to understand our world and ourselves.

In this opening chapter, we'll go over some important preliminaries that we'll need throughout our study. We'll discuss the nature of physical theory and the use of idealized models to represent physical systems. We'll introduce the systems of units used to describe physical quantities and discuss ways to describe the accuracy of a number. We'll look at examples of problems for which we can't (or don't want to) find a precise answer, but for which rough estimates can be useful and interesting. Finally, we'll study several aspects of vectors and vector algebra. Vectors will be needed throughout our study of physics to describe and analyze physical quantities, such as velocity and force, that have direction as well as magnitude.

1.1 The Nature of Physics

Physics is an experimental science. Physicists observe the phenomena of nature and try to find patterns and principles that relate these phenomena. These patterns are called physical theories or, when they are very well established and of broad use, physical laws or principles.

CAUTION: The meaning of the word "theory". Calling an idea a theory does not mean that it's just a random thought or an unproven concept. Rather, a theory is an explanation of natural phenomena based on observation and accepted fundamental principles. An example is the well-established theory of biological evolution, which is the result of extensive research and observation by generations of biologists.

The development of physical theory requires creativity at every stage. The physicist has to learn to ask appropriate questions, design experiments to try to answer the questions, and draw appropriate conclusions from the results. Figure 1.1 shows two famous experimental facilities.

Legend has it that Galileo Galilei (1564–1642) dropped light and heavy objects from the top of the Leaning Tower of Pisa (Fig. 1.1a) to find out whether their rates of fall were

the same or different. Galileo recognized that only experimental investigation could answer this question. From examining the results of his experiments (which were actually much more sophisticated than in the legend), he made the inductive leap to the principle, or theory, that the acceleration of a falling body is independent of its weight.

The development of physical theories such as Galileo's is always a two-way process that starts and ends with observations or experiments. This development often takes an indirect path, with blind alleys, wrong guesses, and the discarding of unsuccessful theories in favor of more promising ones. Physics is not simply a collection of facts and principles; it is also the process by which we arrive at general principles that describe how the physical universe behaves.

No theory is ever regarded as the final or ultimate truth. The possibility always exists that new observations will require that a theory be revised or discarded. It is in the nature of physical theory that we can disprove a theory by finding behaviour that is inconsistent with it, but we can never prove that a theory is always correct.

Getting back to Galileo, suppose we drop a feather and a cannonball. They certainly do not fall at the same rate. This does not mean that Galileo was wrong; it means that his theory was incomplete. If we drop the feather and the cannonball in a vacuum to eliminate the effects of the air, then they do fall at the same rate. Galileo's theory has a range of validity: It applies only to objects for which the force exerted by the air (due to air resistance and buoyancy) is much less than the weight. Objects like feathers or parachutes are clearly outside this range.

Every physical theory has a range of validity outside of which it is not applicable. Often a new development in physics extends a principle's range of validity. Galileo's analysis of falling bodies was greatly extended half a century later by Newton's laws of motion and law of gravitation.

1.2 Solving Physics Problems

At some point in their studies, almost all physics students find themselves thinking, "I understand the concepts, but I just can't solve the problems." But in physics, truly understanding a concept or principle is the same thing as being able to apply it to a variety of practical problems. Learning how to solve problems is absolutely essential; you don't know physics unless you can do physics.

How do you learn to solve physics problems? In every chapter of this book you will find Problem-Solving Strategies that offer techniques for setting up and solving problems efficiently and accurately. Following each Problem-Solving Strategy are one or more worked Examples that show these techniques in action. (The Problem-Solving Strategies will also steer you away from some incorrect techniques that you may be tempted to use.) You'll also find additional examples that aren't associated with a particular Problem-Solving Strategy. Study these strategies and examples carefully, and work through each example for yourself on a piece of paper.

Different techniques are useful for solving different kinds of physics problems, which is why this book offers dozens of Problem-Solving Strategies. No matter what kind of problem you're dealing with, however, there are certain key steps that you'll always follow. (These same steps are equally useful for problems in math, engineering, chemistry, and many other fields.) In this book we've organized these steps into four stages of solving a problem.

All of the Problem-Solving Strategies and Examples in this book will follow these four steps. (In some cases we will combine the first two or three steps.) We encourage you to follow these same steps when you solve problems yourself. You may find it useful to remember the acronym I SEE —short for Identify, Set up, Execute, and Evaluate.

Problem-Solving Strategy

Solving Physics Problems

IDENTIFY the relevant concepts: First, decide which physics ideas are relevant to the problem. Although this step doesn't involve any calculations, it's sometimes the most challenging part of solving the problem. Don't skip over this step, though; choosing the wrong approach at the beginning can make the problem more difficult than it has to be, or even lead you to an incorrect answer.

At this stage you must also identify the target variable of the problem—that is, the quantity whose value you're trying to find. It could be the speed at which a projectile hits the ground, the intensity of a sound made by a siren, or the size of an image made by a lens. (Sometimes the goal will be to find a mathematical expression rather than a numerical value. Sometimes, too, the problem will have more than one target variable.) The target variable is the goal of the problem-solving process; don't lose sight of this goal as you work through the solution.

SET UP the problem: Based on the concepts you selected in the Identify step, choose the equations that you'll use to solve the problem and decide how you'll use them. If appropriate, draw a sketch of the situation described in the problem.

EXECUTE the solution: In this step, you "do the math." Before you launch into a flurry of calculations, make a list of all known and unknown quantities, and note which are the target variable or variables. Then solve the equations for the unknowns.

EVALUATE your answer: The goal of physics problem solving isn't just to get a number or a formula; it's to achieve better understanding. That means you must examine your answer to see what it's telling you. Be sure to ask yourself, "Does this answer make sense?" If your target variable was the radius of the earth and your answer is 6.38 centimeters (or if your answer is a negative number!), something went wrong in your problem-solving process. Go back and check your work, and revise your solution as necessary.

Resuelva las siguientes consignas:

A. Lea la introducción al capítulo y la sección 1.1 y conteste:

1. ¿Qué relación existe entre la física y los ingenieros?
2. ¿Qué es una teoría?
3. Explique la teoría de Galileo Galilei y cómo llegó a esta.
4. ¿Las teorías representan una verdad absoluta? Justifique su respuesta.
5. ¿Quién amplió la teoría de Galileo? ¿De qué manera?
6. En el 1er párrafo de la Sección 1.2 se encuentra el conector **UNLESS**. ¿Qué relación lógica muestra entre los conceptos que une? Mencione cuáles son los dos conceptos.

B. Observe las siguientes oraciones extraídas del texto, qué expresan y cómo las traduciría:

1. Galileo's analysis of falling bodies **was greatly extended** half a century later **by** Newton's laws of motion and law of gravitation.
2. Light and heavy objects **were dropped by** Galileo Galilei.
3. Theories **are** usually **revised or discarded**.
4. The phenomena of nature **are observed** by physicists.

C. Observe las siguientes ideas del texto: qué expresan y cómo las traduciría:

- Identify the relevant concepts.

- Set up the problem

- Execute the solution

- Don't skip over this step

The role of Technology in Today's World and in the Future

Source: <http://techinfopluz.blogspot.com.es/2011/02/robots-and-human-future-world.html>



Technology is an essential part of our lives today and few can imagine living without. We achieved a lot with the help of technology, for example we have the possibility to travel, keep in touch with friends on the other side of the earth and cure many illnesses. It means more freedom and choices for people but at the same time we have to consider the social imbalance, weapons of

mass destruction and natural resource depletion. Jane Godall asks for a reason: *"We are the most intelligent species walking on earth, how it comes we destroy on what we depend?"* What has achieved so far is irrevocable, but we can still determine where it goes in the future.

One of the most important things everybody has to deal with is to overcome the short-termism, which prevails in governments, companies and individuals. Additionally, we know, that decisions one individual takes are not necessarily good for the society as a whole. This doesn't make it easier.

I think it is important to bear in mind the mutual goal: to keep the planet with mankind and all its beautiful animals and plants alive. This implies to make technology as positive as possible for humanity and the environment in the long run. The technology we create now creates the future and it should be for the people and make things easier and not more complicated. So what kind of society will we be?

Technology shapes the future and it can help to make it compatible with nature. It can help us to develop clean energy, transport possibilities with less emissions and low-energy houses to save resources. Technology is not only about technology itself or more efficiency and discovering new methods and processes; we have to add the component of art which is about to make wise choices for the future of technology. We should not develop technology because of itself, but to develop it, because it adds value to society and simplifies human life. For this, it needs to be discussed with experts from many fields and we have to relate the technology to the organization and the culture in which it should operate. And remember: There are no side effects – they are part of the technology.

TED TALKS WITH FURTHER INTERESTING INFORMATION (LAST ACCESSED 30.06.2012). MARIE GLÜCK

TEXT0 14 – GLASDON NEXUS RECYCLING CONTAINERS



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Facilities Manager
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Buenos Aires
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Argentina

Glasdon Nexus® Recycling Containers...

... Helping you to recycle

Many organisations are now implementing initiatives to comply with ISO:14001 Environmental Management Systems, to help minimise the costly and environmental harmful disposal of unsegregated waste. Glasdon supply an extensive range of containers to provide ideal solutions for any recycling scheme. Amongst our products are the following models.

NEW

Nexus 30 - 30 Litre

The Nexus 30 has been designed for areas where space is limited. This aesthetically pleasing 30 litre recycling bin is ideal for desk side and office requirements. Also available as a c-thru model and a deluxe model which offers a contemporary metallic finish.

NEW



Nexus 30

Nexus 50 - 50 Litre

The sleek contemporary design of the Nexus 50 is perfect for establishing reliable recycling schemes in office environments, helping to encourage the efficient collection and segregation of recyclable waste. Manufactured from recycled materials.



Nexus 50

Nexus 100 - 100 Litre

Nexus 100 is our best selling solution for departmental waste segregation schemes. Nexus 100 is used by some of the worlds largest companies to improve their environmental performance. Nexus 100 provides excellent flexibility to meet diverse waste separation requirements. Manufactured from recycled materials.

Glasdon products are supplied world wide either directly from our factory or through our experienced network of distributors.

If you require further information or details of your nearest distributor, please contact us by telephone +44 1253 600435 or by e-mail: sales@glasdon.com

Yours sincerely,

Adrian Steen
Export Manager



More information
over the page



Nexus 100

Resuelva las siguientes consignas:

1. Observe los elementos paratextuales y determine:

- ¿De qué fuente textual se trata?
- ¿Quién es el remitente? ¿Qué área de la empresa envía el documento?
- ¿Quién es el destinatario? ¿Qué área de la institución recibe el documento? ¿Qué relación encuentra entre los productos publicitados y el destinatario?
- Si necesitamos más información ¿cuántas maneras tenemos de conseguir esa información?
- ¿Qué ofrece la empresa Glasdon Nexus?

2. Lea y responda:

- Explique qué es ISO: 14001 y para qué se creó.
- Complete la información faltante en el siguiente cuadro sobre los productos publicitados:

Características	Nexus 30	Nexus 50	Nexus 100
Capacidad	30 litros		
Ideal para colocar en		Oficinas	
Modelos		-----	-----
Características estéticas			
Otras características			Flexibilidad de requerimientos

3. Busque los siguientes conceptos, contextualícelos y determine la función de cada uno:

Recycle - recycled - recycling - recyclable

4. ¿Cuál considera que es el modelo más ecológico de los tres?

TEXT 15 - ARITHMETIC PUZZLES

ENTERTAINING MATHEMATICAL PUZZLES

Martin Gardner

DOVER PUBLICATIONS, INC. NEW YORK

ARITHMETIC PUZZLES

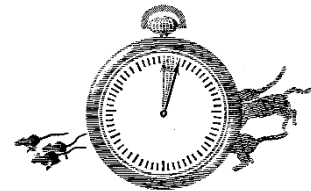
THE NUMBERS THAT are used in counting (1, 2, 3, 4...) are called integers. Arithmetic is the study of integers with respect to what are known as the four fundamental operations of arithmetic: addition, subtraction, multiplication, and division. (Lewis Carroll's Mock Turtle, you may remember, called them Ambition, Distraction, Uglification, and Derision.) Arithmetic also includes the operations of raising a number to a higher power (multiplying it by itself a certain number of times), and of extracting a root (finding a number which, when multiplied by itself a certain number of times, will equal a given number).

It goes without saying that you will never be able to learn algebra or any higher branch of mathematics without knowing your arithmetic well. But even if you never learn algebra, you will find that arithmetic is essential to almost every profession you can think of. A waitress has to add the items on a check, a farmer has to calculate the yield of his crops. Even a shoeshine boy must be able to make change correctly, and making change is pure arithmetic. It is as important in daily life as knowing how to tie your shoelaces.

Puzzles call for nothing more than the ability to do simple arithmetic; and to think clearly about what you are doing.

The Three cats

IF THREE CATS catch three rats in three minutes, how many cats will catch 100 rats in 100 minutes?



MONEY PUZZLES

"IF YOU'LL GIVE me your water pistol," says little Tommy to his playmate, "I'll let you have my dump truck." This kind of trading is called "bartering." In primitive societies it is the only way in which things can be "bought" and "sold."

Think about it a moment and you will see what a poor system this is. A man who wants to sell his cow and buy a horse will be unable to do so until he happens to meet another man who wants to sell his horse and buy a cow. It may be years before he finds such a man. And suppose a man wants to trade his cow for a sheep that belongs to one friend and a pig that belongs to another. He can't slice his cow in half and trade each half separately! So you see, in any complicated society where many things are bought and sold, it is necessary to have something called money; something that can be split up into any amount one wishes, and which has a value that everyone can agree on.

Almost anything can and has been used in the past for money, but today money consists either of coins made of metal or printed paper money. Few uses of arithmetic are more important than learning how to handle money problems. The next five puzzles will test your ability along these lines, and perhaps teach you a few things you did not fully understand before.

Pick your Pay

Suppose you take a new job and the boss offers you a choice between:

(A) \$4,000 for your first year of work, and a raise of \$800 for each year thereafter;

(B) \$2,000 for your first six months of work, and a raise of \$200 every six months thereafter.

Which offer would you take and why?



SECTION 8: GENERAL LABORATORY PRECAUTIONS

8.1 PERSONAL PROTECTIVE EQUIPMENT

The Personal Protective Equipment (PPE) at Work Regulations 1992 require that where PPE is supplied it must have been assessed to ensure it is suitable for the intended purpose.

The appropriate local regulations should be consulted. In many laboratories the wearing of approved types of laboratory coats and safety spectacles is compulsory. The occupants of the laboratory should ensure that any visitors or tradesmen are similarly attired. In some laboratories additional items of personal protective equipment (e.g. face-shields, dust masks, gloves, plastic aprons, etc.) are also provided; the protection worn must be appropriate to the work being undertaken.

8.2 PROTECTIVE CLOTHING AND FOOTWEAR

Bare skin presents areas for contamination by chemicals, micro-organisms and radioactive substances etc.

The laboratory coat provides protection for the arms and body. The choice of style and fabric of lab coats provided must be arrived at as a result of risk assessment of the nature and quantities of the hazards involved and the circumstances of use. The Safety Office has issued guidance to assist in making the correct choice which can be found on the [Safety Office website](#).

Laboratory coats may become contaminated with toxic materials and should normally be left behind when leaving the work area. They must not be worn in any area set aside for tea and coffee making or for the consumption of food or beverages.

They should be washed regularly and whenever they become contaminated with chemicals.

It is necessary to protect the feet and legs from chemical spillage or from damage from heavy equipment being dropped. Suitable footwear must be worn. The wearing of open-toed shoes or sandals in laboratories is not permitted, as their use makes the feet extremely vulnerable to injury from broken glass, spilt corrosive substances, liquid nitrogen, etc. Loose fitting sandals, especially those with no heel restraint, are not secure and may present a tripping hazard.

Loose long hair can be a danger to personal safety in the laboratory. It is readily ignited (lacquer increases this risk) or it could become trapped in equipment or machinery. Long hair should always be tied back.

8.3 PROTECTIVE GLOVES

Gloves may be required to protect the hands against one or more of a range of different hazards, including chemical, biological, heat, and cold. In general, no one type of glove protects well against more than one type of hazard. This makes it very important to select the correct type of glove for the intended use.

Chemical protective gloves are available in a range of materials including natural rubber, neoprene, nitrile, butyl, PVA, PVC and viton. The degree of protection against chemical permeation depends on the glove material, its thickness and the way it is made. Different materials offer widely ranging resistance to permeation as defined by the “breakthrough time”. Glove suppliers provide information to assist in glove selection. It should be remembered that the liquid may be the solvent for other more hazardous chemicals being used and as such will enable these to diffuse through the glove with the solvent, which may also carry the chemical to into the skin.

Solvents may also adversely affect the physical characteristics of the glove and impair its protective properties, for example by leaching plasticising agents out of the glove causing it to become more brittle, cracking and thus leak.

It is important to check gloves for damage such as holes, cuts and distortion. Reusable ones should be examined for signs of internal contamination.

Gloves which bear the term ‘Examination Gloves’ on the box are intended for use in clinical examinations where the patients are to be protected from contamination by the person examining them. These are termed Category 1 gloves and are suitable only for low risk activities. In a laboratory setting, their value is very limited, because they offer little or no protection to the wearer against either biological or chemical hazards. They are of use only in preventing the work from being contaminated by the wearer.

Category 3 gloves will bear CE mark plus four digits e.g. CE 0121 and will be suitable for protection against biological and chemical hazards for most of the uses in the laboratory.

The University Safety Office has published [more detailed guidance](#) on the selection and use of gloves for protection against hazardous substances.

Gloves worn in the laboratory **must** be removed and placed for disposal before leaving the laboratory suite in which such work is being undertaken. The wearing of potentially contaminated gloves in the corridor, rendering others at risk, is strictly forbidden, unless, as a result of risk assessment, the School has sanctioned alternative procedures (e.g. having a glove free hand to negotiate doors). The normal requirements on removal of protective clothing prior to entering offices, seminar rooms, rest areas and when entering public areas apply.

<https://www.nottingham.ac.uk/safety/handbook/general-laboratory-precautions.aspx>

Resuelva las siguientes consignas

Prelectura

1. Observe la información paratextual y explique qué tipo de texto es.

2. ¿Qué recursos están involucrados en un laboratorio? ¿Cómo es el laboratorio de UTN? ¿Hay uno solo?
3. ¿A quién está dirigido este texto?
4. Entre a la página de internet y compare estos contenidos con los del laboratorio de UTN.

Lectura

5. Observe las frases verbales marcadas en negrita en el texto y explique por qué se utilizan en el texto. Busque otros ejemplos similares.
6. ¿A qué hacen referencia las palabras encuadradas? (those, ones, which)
7. ¿Qué relación lógica muestra el conector **UNLESS** (subrayado en el texto), y cuáles son los dos conceptos que relaciona?
8. Lea el artículo con un compañero y explique las similitudes con lo que se tiene en cuenta en la Facultad.

Post Lectura

9. Realice una síntesis con un compañero sobre la Sección 8.3 incluyendo los conceptos más importantes.

- [News](#)
- [Technology](#)
- [Artificial intelligence \(AI\)](#)

SMART ROBOTS, DRIVERLESS CARS WORK – BUT THEY BRING ETHICAL ISSUES TOO

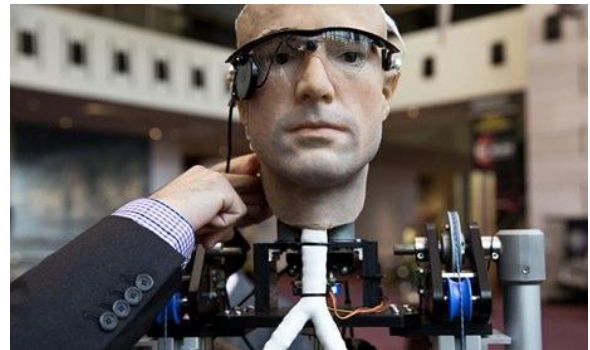
Educational robots, intelligent implants, brain chips: scientists and legal experts will meet this week to debate how to deal with the rapid march of artificial intelligence

Nicola Davis

The Observer, Sunday 20 October 2013

An engineer adjusts the robot "The Incredible Bionic Man", at the Smithsonian National Air and Space Museum in Washington, the world's first-ever functioning bionic man made of prosthetic parts and artificial organ implants.

Photograph: Joshua Roberts/REUTERS



From personalised searches of Google to the seductive experience of driverless cars, from educational robots that hone your French to prosthetics that are stronger and faster than our own limbs: artificial intelligence is poised to revolutionise our lives.

Now scientists, legal experts and philosophers are joining forces to scrutinise the promise of intelligent systems and wrangle over their implications. This week in Brighton, the fourth EuCogIII members' conference is set to tackle these issues head on. "Fundamentally we're interested in considering the ethical and societal impact of such systems," says Alan Winfield, professor of electronic engineering at UWE Bristol. It is time, he says, to make some crucial decisions. "If we get it wrong, there are consequences right now."

It's a point well illustrated by IBM's intelligent system, Watson. Two years after thrashing human contestants at the quickfire quiz *Jeopardy!*, Watson has graduated from game shows to medical school and could soon be diagnosing diseases. This year commercial products based on Watson were unveiled for clinical use, harnessing the system's ability to crunch through swaths of medical information and make decisions.

"There is a huge amount of knowledge now that doctors can potentially have. Obviously they can't absorb all of it and they can't necessarily remember all of it," says Tony Prescott, professor of cognitive neuroscience at the University of Sheffield. With access to the latest developments as well as the medical records of patients, systems such as Watson could suggest an accurate diagnosis faster and more often, as well as predicting an individual's health risks.

But there is a hitch. With intelligent systems accessing medical records comes the fear of compromised privacy and security, as many will be connected via the internet. Could we, or even should we, be allowed to opt out of such an intelligent system? "It is a decision we have to make as a

society," says Prescott. "Whether we want to give up some of our privacy in order to get improved services like better healthcare."

But how far can we trust such systems? Putting your faith in a "black box" may seem at best naive, at worst reckless. It is an issue that boils down to trust, making it essential that doctors are closely involved in training such systems, understanding how they work and confirming the diagnoses are spot on.

At the heart of the revolution is you, the consumer. With computers getting smaller, more powerful and more energy-efficient, few areas of our lives will remain untouched by intelligent machines. Driverless cars are expected to cause a storm. "The technology is ready," says Winfield. "The problem is insurance and legislation." While driverless cars could offer many benefits, from bringing independence to the elderly to reducing the number of road accidents, disasters could still happen. Who then pays the damages – the owner, or the car producer?

Last year a European research project, RoboLaw, was created to tackle such legal conundrums and will deliver its guidelines on regulations to the European commission in the spring. One question is whether it's time to rethink liability to ensure safety and justice without compromising the incentive for companies to develop the technology – "for instance, through the usage of compulsory insurance schemes or by assessing so-called 'safe harbours' to shield, in some cases under certain conditions, the liability of the producer of the car," explains Andrea Bertolini, a post-doctoral fellow in private law at the Scuola Superiore Sant'Anna in northern Italy and a member of the RoboLaw team.

And it is not just issues of liability that could be reformed. Fallible humans are constrained by speed limits to reduce the number of crashes, but with an all-encompassing knowledge of road layout and road users, intelligent cars could themselves decide how fast they travel, banishing the need for fixed limits.

One of the greatest issues, says Bertolini, is that there are many types of robots each posing different legal problems. State of the art prosthetic devices – essentially wearable intelligent robots – could soon outperform our natural limbs, raising new concerns that the technology could become available to individuals who may wish to trade in their healthy body parts for a prosthesis. "Should this be regulated, and eventually if it should be regulated, how should it be regulated?" asks Bertolini.

The questions become even more pressing when the possibility of implants are considered – imagine a brain chip that could let you check your email, search the internet or tap in to GPS. It's the ultimate "hands-free" device.

This possibility of becoming "bio-hybrid" may sound futuristic, unlikely even. But when technology develops, it develops quickly. "It is moving way faster than legislation can keep up and yes, it's a problem," says Tony Belpaeme, professor of cognitive systems and robotics at the University of Plymouth.

And the issues are international. "The trouble is that your data is now globally spread and legislation isn't the same across various regions across the planet," says Belpaeme. With recent revelations over the access and use of data by various government agencies still reverberating, issues of data storage, privacy and security need to be aired openly. "How much worse would it be if there were such external and covert constraints on cognitive technology?" asks Dr Ron Chrisley, reader in philosophy at the University of Sussex and one of the conference organisers. With the possibility of technology becoming intertwined with our very bodies, the threat of unauthorised access looms large.

Running scared is not an option. Intelligent systems offer us the chance to hone many fundamental areas of our lives, including education. It's an effect Belpaeme has seen firsthand through his research into use of social robots in education. While current computer systems can support learning, robots, particularly those sporting a face and personalised conversation, evoke a stronger response. "What

we found is that if you have a robot there taking you through exactly the same exercises, the children learn faster and better," says Belpaeme.

"I can just see a future where you have one or two robots sitting in the corner of a classroom," he says. "If you just need a little push or you want to be challenged, you get 20 minutes or half an hour with a robot." According to Belpaeme, hospitals could benefit from such technology, with robots teaching children how to manage their medical conditions, while in care homes such robots could help the elderly with their daily exercises.

With such technological leaps set to transform our lives we, the public, need to be involved in the discussion, shaping policy and priorities from the outset. "I think that the greatest risk with these kinds of technologies is that they come along and they are a big surprise to people," says Prescott. Which is why, before the conference, you are invited to post questions for experts to discuss. You can also follow the event on Twitter through the hashtag #robotsandyou.

Intelligent machines could turn education, healthcare and daily life into optimised, tailored experiences. Getting society on side is big, and it's clever.

FUTURE QUESTIONS

Driverless cars

- Who pays damages in an accident?
- Should children and the elderly be allowed to 'drive' them too?
- Should such cars have to obey current road rules, such as speed limits?

Intelligent prostheses

- Will robotic enhancements give users an unfair advantage?
- Should intelligent prostheses be available to the able-bodied?
- Who decides whether you should be allowed a prosthesis?

Social robots

- Could faults in or hacking of intelligent robots cause physical harm to users?
- Will intelligent robots redefine the roles of teachers, carers and others?

Implanted devices

- Will intelligent implants mean the end of privacy?
- Who will technically own the device and its data?

Medical assistants

- How far should doctors trust intelligent systems?
- Who gives consent to sharing medical records with intelligent systems?

-----**Resuelva las siguientes consignas en español**

Prelectura

1. Complete la siguiente información: título / autor / fecha / fuente de publicación
2. De acuerdo al título, imagen y epígrafe, ¿Cuál es el tema del artículo?
3. Lea el copete y responda: cuándo, quiénes, por qué y para qué.
4. El título muestra un contraste. ¿Qué podría decir el texto al respecto?

Lectura

5. Lea la 1ra oración de los 5 primeros párrafos. ¿Qué información nueva puede agregar?
6. De acuerdo al texto, hay un encuentro de profesionales:
 - Cómo se llama el encuentro
 - De qué áreas son los profesionales
 - Dónde se realiza
 - Cuál es el tema a desarrollar
7. Escanee el texto completo y responda en qué áreas se están utilizando robots.
8. Explique con sus palabras a qué se refiere el autor con la frase "*At the heart of the revolution is you, the consumer*".
9. Encuentre la opinión de un experto que describa ventajas acerca del avance de la IA, y, por otro lado, que describa problemas respecto del mismo tema.
10. ¿En qué párrafos el autor hace referencia a los problemas éticos con respecto a este tema?
11. Busque dos oraciones en el texto que muestren la opinión del autor sobre el avance de la IA.



[Privacy and Security Notice](#)

Questions and Answers

[| Return to the Electricity and Magnetism Index Page |](#)



How do I make an electromagnet?

It is fairly easy to build an electromagnet. All you need to do is wrap some insulated copper wire around an iron core. If you attach a battery to the wire, an electric current will begin to flow and the iron core will become magnetized. When the battery is disconnected, the iron core will lose its magnetism. Follow these steps if you like to build the electromagnet described in our Magnets and Electromagnets experiment:

Step 1 - Gather the Materials

To build the electromagnet described in our Magnets and Electromagnets experiment, you will need:

One iron nail fifteen centimeters (6 in) long

Three meters (10 ft) of 22 gauge insulated, stranded copper wire

One or more D-cell batteries

A pair of wire strippers

Step 2 - Remove some Insulation

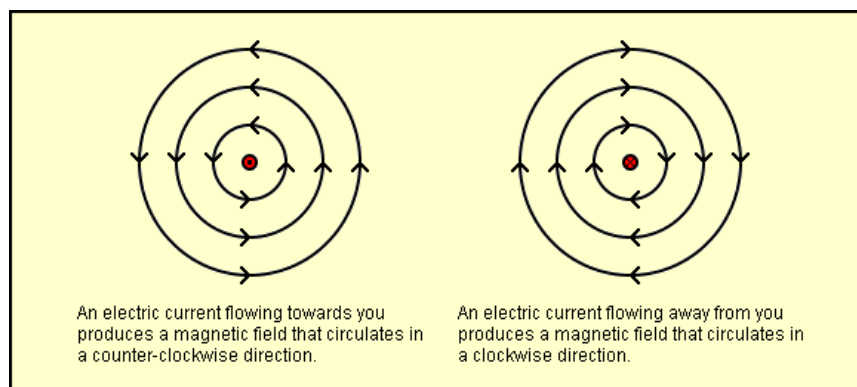
Some of the copper wire needs to be exposed so that the battery can make a good electrical connection. Use a pair of wire strippers to remove a few centimeters of insulation from each end of the wire.

Step 3 - Wrap the Wire around the Nail

Neatly wrap the wire around the nail. The more wire you wrap around the nail, the stronger your electromagnet will be. Make certain that you leave enough of the wire unwound so that you can attach the battery.



When you wrap the wire around the nail, make certain that you wrap the wire all in one direction. You need to do this because the direction of a magnet field depends on the direction of the electric current creating it. The movement of electric charges creates a magnetic field. If you could see the magnetic field around a wire that has electricity flowing through it, it would look like a series of circles around the wire. If an electric current is flowing directly towards you, the magnetic field created by it circles around the wire in a counter-clockwise direction. If the direction of the electric current is reversed, the magnetic field reverses also and circles the wire in a clockwise direction. If you wrap some of the wire around the nail in one direction and some of the wire in the other direction, the magnetic fields from the different sections fight each other and cancel out, reducing the strength of your magnet.



Step 4 - Connect the Battery

Attach one end of the wire to the positive terminal of the battery and the other end of the wire to the negative terminal of the battery. If all has gone well, your electromagnet is now working!

Don't worry about which end of the wire you attach to the positive terminal of the battery and which one you attach to the negative terminal. Your magnet will work just as well either way. What will change is your magnet's polarity. One end of your magnet will be its north pole and the other end will be its south pole. Reversing the way the battery is connected will reverse the poles of your electromagnet.

Hints to Make Your Electromagnet Stronger

The more turns of wire your magnet has, the better. Keep in mind that the farther the wire is from the core, the less effective it will be.

The more current passes through the wire, the better. **Caution!** Too much current can be dangerous! As electricity passes through a wire, some energy is lost as heat. The more current that flows through a wire, the more heat is generated. If you double the current passing through a wire, the heat generated will increase **4 times!** If you triple the current passing through a wire, the heat generated will increase **9 times!** Things can quickly become too hot to handle.

Try experimenting with different cores. A thicker core might make a more powerful magnet. Just make certain that the material you choose can be magnetized. You can test your core

with a permanent magnet. If a permanent magnet is not attracted to your core, it will not make a good electromagnet. An aluminum bar, for example, is not a good choice for your magnet's core.

This page is maintained by Steve Gagnon.

Resuelva las siguientes consignas

A) Lea el texto y responda las siguientes preguntas:

- 1) ¿Qué materiales se necesitan para hacer un electroimán?
- 2) ¿Cómo se fabrica uno?
- 3) ¿Por qué es importante bobinar el cable en una misma dirección?
- 4) ¿Es indistinto conectar un extremo del cable al terminal positivo o negativo de la pila?
¿Existe alguna diferencia?
- 5) ¿Qué se debe hacer para obtener un electroimán más fuerte?

B) Observe las siguientes oraciones extraídas del texto ¿Qué expresan? ¿Cómo las traduciría?

- The more** wire you wrap around the nail, **the stronger** your electromagnet will be.
- The more** turns of wire your magnet has, **the better**.
- Keep in mind that **the farther** the wire is from the core, **the less effective** it will be.
- The more current** passes through the wire, **the better**.

1. If the direction of the electric current is reversed, the magnetic field reverses also and circles the wire in a clockwise direction.
2. If you wrap some of the wire around the nail in one direction and some of the wire in the other direction, the magnetic fields from the different sections fight each other and cancel out, reducing the strength of your magnet.
3. If you attach a battery to the wire, an electric current will begin to flow and the iron core will become magnetized.
4. When you wrap the wire around the nail, make certain that you wrap the wire all in one direction.
5. If you could see the magnetic field around a wire that has electricity flowing through it, it would look like a series of circles around the wire.



Paper Recycling

Australian homes and offices use millions of tonnes of paper per year for printing, writing and reading alone. Recycling paper products and I turn purchasing recycled products has environmental and economic benefits.

Reducing Paper Waste

There are many ways to reduce your paper use:

- Think before printing – do as much as you can on a screen
- Use both sides of a sheet of paper – default printing to double-sided, re-use the reverse side of single-printed pages
- Avoid junk mail by using a 'No Advertising Material' sticker



Recycling paper reduces energy, water and natural resource use

Why Recycle

When paper breaks down in landfill it creates methane, a major greenhouse gas with the global warming capacity 21 times more powerful than carbon dioxide. Recycled paper uses up to 50% less energy and 90% less water than making it from raw materials.

Where possible, purchase paper products with recycled paper content of up to 100% to close the recycling loop.

How to Recycle

Paper is the most commonly collected material in kerbside collections.

- Keep paper free of food scraps and plastic, which can contaminate the process of recycling
- Used tissues and paper that has been in contact with food scraps can be recycled in a worm farm, Bokashi bucket or compost heap or sent put in the rubbish
- Workplaces can keep high quality office paper in circulation by collecting white office paper separately from newsprint, coloured paper and cardboard so it can be recycled into new office paper.

Resuelva las siguientes consignas:

a. Observe la organización del siguiente texto, sus componentes textuales o lingüísticos y paratextuales y conteste:

1. ¿Qué tipo de texto es?
3. ¿Propósito del texto?

2. Género textual:
4. ¿Cuál es el tema del texto?

b. Lectura global:

¿Cómo está organizado el texto?

¿Qué función cumple la primera oración de cada subtítulo? ¿Qué informan?

What Happens to the Paper?

Paper is broken down chemically or mechanically to free the fibres and create pulp. The pulp is run through special rollers that remove the moisture and flatten material into giant rolls of paper. Waste products from the recycling process, ink, short fibres and plastics (called 'sludge') are either sent into landfill, burnt for energy or used as soil conditioner.

Choosing Australian made recycled paper help support local green jobs and contributed to the economy. Australian Paper manufacturers brands such as Planet Ark Paper and Reflex in the La Trobe Valley of Victoria while ABC Tissue makes the Naturele range of toilet tissue in Brisbane. Recycled paper products include: office paper, packaging, toilet paper, egg cartons, soundproofing, furniture and cardboard. Paper can be recycled between 6 and 8 times with the quality reducing each time (as the fibres become shorter).



55,000 tonnes of office paper a year would be saved if every Australian household used Naturele Toilet Tissue in just one bathroom



Planet Ark Paper is made with 100% recycled Australian waste paper, is carbon neutral and FSC certified

Take Action

- Visit RecyclingNearYou.com.au to find out if your local council accepts paper for recycling and other drop-off locations
- Buy recycled paper products such as [Planet Ark Paper](#) and [Naturele Tissue](#)
- For businesses find paper recycling services on our [Business Recycling](#) website

Further information

To find out about cardboard recycling see below.

[Cardboard Factsheet](#)
[Business Recycling - Paper and Cardboard](#)

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1. ¿Qué se puede hacer para reducir los desechos de papel?
2. ¿Por qué es importante reciclar el papel?
3. ¿Cuáles son las ventajas mencionadas al reciclar papel?
4. ¿Qué pasos previos al reciclaje sugieren realizar al lector para colaborar desde sus hogares?
5. Explique cómo es el proceso de reciclaje.

Friendly Machines

Making human-friendly robots is a pressing challenge and a big opportunity.

April 23, 2013



As a research scientist I hear this question all the time: when are your robots going to replace me? But that is certainly not my goal.

A more important objective, to my mind, is making robots more human-friendly in their form, behavior, and function. By this I mean that robots should be appealing and approachable. They should behave in ways that are easy for humans to interpret, and they should perform functions that meet human needs. This applies in places like factories where more robots can work effectively alongside people (see “Baxter: The Blue-Collar Robot,”). This is not about making human-like robots. Humanoid robots have a place in entertainment, medical training, and possibly other domains, but human-friendly robots are not necessarily humanoid. In fact, by setting user expectations too high, looking too human could make it more difficult for a robot to interact with people. We are often disappointed and frustrated with the limited capabilities of robots that look as if they should be just as smart as we are.

These robots also do not need to behave just like humans. They might, for example, behave more like service dogs. As long as they are predictable, robots have a hope of making it in the everyday world. Many people know how to communicate with dogs just fine without needing language at all.

Finally, these human-friendly robots must meet real human needs, not only the needs of their inventors. Fetch-a-beer and fold-a-towel demos are nice scientific steps toward building more general robotic capabilities. But what we need now is for human-centered-design researchers and product-minded entrepreneurs to do the dance of the necessary and the possible with the robotics community. Why does this humanist stuff matter? Because it will help us realize the true potential of the technology. Too many long-term studies of robots in hospitals, offices, and homes have revealed the problem with ignoring the importance of human-to-robot interaction: the robots end up interred in closets, retired to garages, or “mysteriously” disabled and shoved under desks.

Many of my robotics colleagues cringe at the challenges presented by unstructured environments that personal robots need to navigate. But the untrained people around these robots present an entirely different set of equally important challenges. Without serious involvement from the interaction-design, product-design, and entrepreneurial communities, personal robots don’t stand a chance of surviving out in the “real world.”

Resuelva las siguientes consignas

Lectura:

- 1) ¿Cuál es la postura central de la autora en cuanto al diseño de robots?
- 2) ¿Qué posición adopta frente a los robots humanoides? ¿Cómo la justifica?

- 3) ¿Qué otros argumentos esgrime la autora para defender su tesis?
- 4) ¿A qué se refiere con la frase "this humanist stuff"?
- 5) ¿Qué predicción realiza con respecto al uso de robots personales? ¿Cómo se relaciona dicha predicción con su postura?

Post-lectura:

Redacte una síntesis del texto en un párrafo de tres oraciones que incluya las ideas más importantes del texto y el propósito del autor.

History of Engineering



WRITTEN BY: Ralph J. Smith

The first engineer known by name and achievement is Imhotep, builder of the Step Pyramid at Şaqqārah, Egypt, probably in about 2550 BC. Imhotep's successors—Egyptian, Persian, Greek, and Roman—carried civil engineering to remarkable heights on the basis of empirical methods aided by arithmetic, geometry, and a smattering of physical science. The Pharos (lighthouse) of Alexandria, Solomon's Temple in Jerusalem, the Colosseum in Rome, the Persian and Roman road systems, the Pont du Gard aqueduct in France, and many other large structures, some of which endure to this day, testify to their skill, imagination, and daring. Of many treatises written by them, one in particular survives to provide a picture of engineering education and practice in classical times: Vitruvius' *De architectura*, published in Rome in the 1st century AD, a 10-volume work covering building materials, construction methods, hydraulics, measurement, and town planning.

In construction medieval European engineers carried technique, in the form of the Gothic arch and flying buttress, to a height unknown to the Romans. The sketchbook of the 13th-century French engineer Villard de Honnecourt reveals a wide knowledge of mathematics, geometry, natural and physical science, and draftsmanship.

In Asia, engineering had a separate but very similar development, with more and more sophisticated techniques of construction, hydraulics, and metallurgy helping to create advanced civilizations such as the Mongol empire, whose large, beautiful cities impressed Marco Polo in the 13th century.

Civil engineering emerged as a separate discipline in the 18th century, when the first professional societies and schools of engineering were founded. Civil engineers of the 19th century built structures of all kinds, designed water-supply and sanitation systems, laid out railroad and highway networks, and planned cities. England and Scotland were the birthplace of mechanical engineering, as a derivation of the inventions of the Scottish engineer James Watt and the textile machinists of the Industrial Revolution. The development of the British machine-tool industry gave tremendous impetus to the study of mechanical engineering both in Britain and abroad.

The growth of knowledge of electricity—from Alessandro Volta's original electric cell of 1800 through the experiments of Michael Faraday and others, culminating in 1872 in the Gramme dynamo and electric motor (named after the Belgian Z.T. Gramme)—led to the development of electrical and electronics engineering. The electronics aspect became prominent through the work of such scientists as James Clerk Maxwell of Britain and Heinrich Hertz of Germany in the late 19th century. Major advances came with the development of the vacuum tube by Lee De Forest of the United States in the early 20th century and the invention of the transistor in the mid-20th century. In the late 20th century electrical and electronics engineers outnumbered all others in the world.

Chemical engineering grew out of the 19th-century proliferation of industrial processes involving chemical reactions in metallurgy, food, textiles, and many other areas. By 1880 the use of chemicals in manufacturing had created an industry whose function was the mass production of chemicals. The design and operation of the plants of this industry became a function of the chemical engineer.

Source: <https://www.britannica.com/technology/engineering>

Inequality and innovation as seen from the South

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Abstract

This article begins by contrasting different approaches to Science, Technology and Society (STS), and suggests that an interactionist model is best for understanding development issues. It then argues that in a perspective “from the South”, interactionism points as well toward adaptation of a national systems of innovation (NSI) approach and the need for a more careful analysis of the features of underdevelopment that are characteristic of specific countries in the age of globalization. Following a review of NSI theory, the article focuses on some relations between inequality, capabilities, and technical change, with comparisons between development trajectories in Scandinavia and the Southern Cone of the Americas. “Learning divides” are characterized as a core dimension of underdevelopment today. Historical comparisons suggest a distinction between proactive and reactive types of equality as important for development studies. A conclusion presents the search for “proactive equality” as a fundamental task for STS studies in underdeveloped contexts.

In the STS studies, there are two main approaches: a “European” tradition that focuses on the social factors shaping the generation of new science and technology outputs, and a [north] “American” tradition that focuses on the social consequences of such outputs [1] (p. 66). But another tradition defined by J.D. Bernal in his great work on social history of science [2] studies the interactions of scientific and technological practices with all social relations. This approach has a long if currently unappreciated history, especially in the South, where a remarkable example is provided by Latin American thinking of the 1960s and 1970s on the “science- technology-development-dependency” issues [3] (esp. Sec. III). This “third tradition” inspires our “innovation as seen from the South” approach that is briefly presented here. It aims to make a small contribution at the crossroads of innovation studies and development problems.

1. Underdevelopment today

Science and technology foster processes of destabilization and change. The ever- increasing economic relevance of knowledge and the immediacy of its applications fuel innovation. The concentration of knowledge resources and innovation capabilities in some groups and regions leads to the self-reproduction of social asymmetries. Production grows and becomes more diversified, as does inequality. Innovation implies “creative destruction,” as Joseph Schumpeter [4] put it, but for many people what is lost is more important than what is gained. In the natural and social environments in which they live, what prevails is “destructive innovation”. With regard to products, machines, jobs, and skills, “creation” for some people often means “destruction” for others.

Today we are witnessing an acceleration of the process of inclusion/exclusion, and even the growing irreversibility of the latter. Almost twenty years ago, Fernando Henrique Cardoso observed:

Part of what has been known as the Third World has become a wide “Fourth” World of need, hunger and despair. The case is not any more that of a periphery tied to a capitalist centre by classic dependency relationships. We are witnessing the dramatic situation of countries that failed in occupying the slightest interstice in the world market and which exploitation has ceased to be in the center’s interest [5] (p. 444).

At the end of the 1990s, in his exhaustive study of “the information age”, Manuel Castells reached similar conclusions [6].

The predominant counter-vision presents globalization as a process from which every place and person will eventually benefit. Exclusionary trends were initially explained as caused by the incomplete application of necessary structural changes. More recently, the negative implications of techno-economic changes are being accepted in international circles, and social policies are once again recommended as responses to the negative side effects of globalization. In the midst of uncertainty, destabilization, and change brought about by knowledge-based and innovation-driven economies, debates tend to be polarized between “apocalypse” and “integration”, to adopt terms from Umberto Eco [7].

Apocalyptic visions are hard to avoid, at least in the South, and not only there. Extreme poverty, dramatic demographic changes due to massive migrations, new forms of violence and urban insecurity, monetary transfers, the great difficulties of coping with open or functional illiteracy—these are just some of the challenges to all positive interpretations of prevailing trends.

Intellectual integration is another attractive option. Today the major criterion of success is determined by international benchmarking. Dominant ideas are based on the material success of advanced capitalism. They exert an attraction of the type described by Gramsci when elaborating the notion of hegemony [8] (p. 487). In such a context it is not easy to put forth an alternative intellectual framework. The hegemonic system sees current trends as fundamentally positive and argues that prevailing policies are applicable worldwide and will eventually yield beneficial results. Thus, is reasserted the starting point of development thinking [9]: once again, it is widely assumed that a unique way to overcome underdevelopment is to follow the North, directly applying concepts and policies elaborated under alien circumstances.

Such has not always been the case. From the 1950s to the 1970s alternative approaches to development issues were highly influential. Two examples from Latin America are “ECLA structuralism” (ECLA stands for the Economic Commission for Latin America) and “dependency theory”. Now widely rejected as naive, their basic assumption nevertheless deserves reconsideration: a specific analysis of the “peripheral condition” is needed. In other words, context matters; the same institutions or policy instruments may play quite different roles in central countries than in peripheral regions.

Our point of departure may **thus** be summarised as follows: (a) global trends are not closing the gap between North and South, so it is necessary to consider alternative policies; (b) knowledge, innovation, and learning are core problems of underdevelopment, so it is not possible to reject prevailing views without specifically addressing such issues; and (c) for revitalizing development thinking, some approaches elaborated in the North can be illuminating if reconsidered in the light of Southern realities.

2. The national systems of innovation (NSI) approach

...

3. Inequality and development

...

4. Learning divides

...

5. Interactions between equality and innovation

...

6. Conclusion: in search of proactive equality

A primary link between growth and inequality seems to be the following: if a period of economic growth does not give place to an important educational effort that diminishes inequality in this fundamental aspect, then further economic growth may be difficult. This has been true at least since the Industrial Revolution, but it is more important today. “Innovation is predominantly a labor-intensive process” [28] (p. 109). Since economic growth is increasingly based on knowledge and led by innovation, it is predominantly a highly qualified labor-intensive process.

The most successful development processes of the twentieth century have many differences between them, but they all show specific combinations of (a) important progress toward less inequality, especially concerning education, (b) virtuous circles between equality, education and a positive technological imaginary, and (c) active and efficient industrial policies, implemented over long periods, and with strong research support.

A simple but non-trivial lesson stems from every story of success concerning long-range development. The combination of learning in a formal system and at work, in a context of relatively low social distance, is a major source of proactive equity, that is, of egalitarian trends that multiply collective capabilities for adapting to changes, for expanding the uses of new technologies, and for upgrading productive activities and innovation. It is difficult to promote equality and even more difficult to promote proactive equality. Development has not been a simple task in the past and does not look easier today.

At the dawn of the twenty-first century, we are witnessing the multiplication and diversification of examples of a vicious cycle in which inequality generates inequality through learning divides. Socio-geographic positions condition the opportunities for study at an advanced level and of learning by working in environments where knowledge is a basic resource. Such opportunities, in turn, are fundamental for acquiring the capabilities that are needed to cope with innovation challenges, to profit from new possibilities, and to face the new dangers that are also being created—all at increasing speeds. The process that intertwines inequality and learning divides is perhaps the essence of the underdevelopment problem today.

Summing up, when approaching innovation from the South, a fundamental issue appears at the top of the research agenda: the empirical and theoretical study of proactive types of equality.

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Rodrigo Arocena holds doctorates in Development Studies and in Mathematics, and is Professor of Science and Development at the University of the Republic, Uruguay. He has widely published on development problems and higher education. His most recent book is La Universidad LatinoAmericana del Futuro (México, 2001), with Judith Sutz.

Judith Sutz is an electrical engineer with a Master’s degree in Developing Planning and a doctorate in Socioeconomics of Development. She serves as Academic Coordinator of the Scientific Research Commission at the University of the Republic, Uruguay. She has published widely on innovation policies and universities.

Resuelva las siguientes consignas

Prelectura

1. ¿Qué información paratextual obtenemos del artículo? ¿Cuál es la información extra sobre los autores?
2. ¿En qué asignatura estudia estos temas?
3. De acuerdo a lo que conocemos hasta aquí, ¿puede predecir de qué tratará el artículo?

Lectura del abstract

4. Lea la 1ra oración de los dos párrafos que componen el abstract y relaciónelas con el título.
5. ¿Qué significan las siglas STS y NSI que están en el abstract?
6. Mencione cuáles son los diferentes enfoques hacia la STS.
7. ¿Cuál es el objetivo del artículo?

Sección 1. Underdevelopment today

8. Lea nuevamente, en esta sección, la 1ra oración de cada párrafo y explique qué información nueva ha obtenido.
9. Explique con sus palabras la frase “the growing irreversibility of the latter”.
10. ¿En qué parte se ve una ampliación del objetivo antes mencionado?
11. En el último párrafo de esta sección se encuentra el conector **THUS**. ¿Qué relación lógica presenta y cuáles son los conceptos que une?

Sección 6. Conclusion

12. Explique la importancia del concepto “learning divides” que se observa a lo largo del artículo.
13. Realice una síntesis en TRES ORACIONES de la conclusión a la que llegan los autores.



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Cloud computing adoption by SMEs in the north east of England

A multi-perspective framework

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 Newcastle upon Tyne, UK, and*

Feng Li
Cass Business School, London, UK

Abstract

Purpose – The purpose of this paper is to contribute to a growing body of research on cloud computing, by studying the small to medium-sized enterprise (SME) adoption process. If SMEs have access to scalable technologies they could potentially deliver products and services that in the past only large enterprises could deliver, flattening the competitive arena.

Design/methodology/approach – By adopting the Technological, Organisational and Environmental (TOE) framework as a theoretical base, this qualitative exploratory study used semi-structured interviews to collect data in 15 different SMEs and service providers in the north east of England. The north east of England was selected as it is a region that aspires to become home to innovative digital firms and most of the companies in the region are SMEs.

Findings – The main factors that were identified as playing a significant role in SME adoption of cloud services were: relative advantage, uncertainty, geo-restriction, compatibility, trialability, size, top management support, prior experience, innovativeness, industry, market scope, supplier efforts and external computing support. In contrast, this study did not find enough evidence that competitive pressure was a significant determinant of cloud computing adoption.

Research limitations/implications – These findings have important implications and great value to the research community, managers and information and communication technologies (ICT) providers, in terms of formulating better strategies for cloud computing adoption. For service providers, using the research model in this study can assist in increasing their understanding of why some SMEs choose to adopt cloud computing services, while seemingly similar ones facing similar market conditions do not. Also, cloud computing providers may need to improve their interaction with SMEs which are involved in the cloud computing experience, in an effort to create a healthy environment for cloud computing adoption, and to remove any vagueness surrounding this type of technology.

Originality/value – This study is an attempt to explore and develop an SME cloud computing adoption model that was theoretically grounded in the TOE framework. By adopting the TOE framework this study has shown that the three contexts of this framework (technological, organisational, and environmental) are connected to each other.

Keywords Information technology, Communication technologies, Innovation, Internet, Information and communication technologies innovation, Cloud computing, North east of England, TOE framework, Small to medium-sized enterprises, Diffusion of innovation

Paper type Research paper

1. Introduction

1 The use of information and communication technologies (ICT) can improve business competitiveness, and has provided genuine advantages for small- and medium-sized enterprises (SMEs, firms with one to 250 employees (DTI, 2004)), enabling them to compete with large firms (Swash, 1998; Bayo-Moriones and Lera-Lopez, 2007). In the UK, SMEs represent the vast majority of all enterprises, 99 per cent according to BIS (2010), a figure which is not atypical for virtually all countries. Despite their importance, limited research has studied the adoption, implementation and usage of ICT in the small enterprises within the UK context (Shiels et al., 2003; Harindranath et al., 2008). In traditional IT environments, increasingly the complex management issue of software, hardware and networking equipment require specialist staff for implementing and maintaining IT services (Thinkstrategies, 2002). Some of the promised benefits from cloud computing can be very appealing for SMEs, which need to maximise the return on their investment and still remain competitive in an ever demanding business environment. Saya et al. (2010) point out that while extant research has studied cloud computing architecture (Rochwerger et al., 2009), potential applications (Liu and Orban, 2008) and costs and benefits (Assuncao et al., 2009), the decision making on the adoption of cloud computing has not been empirically examined.

2 Consequently, this paper's main research objective is to contribute to a growing body of research on cloud computing, by studying the SME adoption process. If SMEs have access to scalable technologies they could potentially deliver products and services that in the past only large enterprises could deliver, flattening the competitive arena. The north-east of England was selected for this study as it is a region that aspires to become home to innovative digital firms and most of the companies in the region are SMEs.

3 In the next section, the relevant literature is presented and the theoretical framework adopted in this study is discussed in detail. This is followed by a description of the study's overall research design and methodological issues. The paper continues by presenting the study's main findings, before highlighting future research avenues.

2. Literature review

- 2.1 ICT innovation adoption
- 2.2 The TOE framework

3. Methodology

- 3.1 Research design
- 3.2 Data collection and analysis
- 3.3 Sampling

4. Findings and discussion

- 4.1 Technological context
- 4.2 Organisational context
- 4.3 Environmental context
- 4.4 Discussion
- 4.5 TOE, cloud computing and SMEs

5. Conclusion

4 “Computing services on-demand” is gradually modifying the way information system services are developed, scaled, maintained and paid for. Previous trends such as virtualisation and outsourcing may be considered as the initiation point for cloud computing ideas. Nevertheless, the novelty of cloud computing is the comprehensive method of providing computing services using high-speed internet connections (Leimeister et al., 2010; Yadav and Zeng Wen, 2010).

5 This study is an early attempt to explore and develop an SME cloud computing adoption model that was theoretically grounded in the TOE framework. By adopting the TOE framework this study has shown that the three contexts of this framework (technological, organisational and environmental) are connected to each other.

6 The main factors that were identified as playing a significant role in SME adoption of cloud services were: relative advantage, uncertainty, geo-restriction, compatibility, trialability, size, top management support, prior experience, innovativeness, industry, market scope, supplier efforts and external computing support. In contrast, this study did not find enough evidence that competitive pressure was a significant determinant of cloud computing adoption.

7 **These findings** have important implications and great value to the research community, managers and ICT providers, in terms of formulating better strategies for cloud computing adoption. For service providers, using the research model in this study can assist in increasing their understanding of why some SMEs choose to adopt cloud computing services, while seemingly similar ones facing similar market conditions do not. **On the other hand, however**, cloud computing providers may need to improve their interaction with SMEs who are involved in the cloud computing experience, in an effort to create a healthy environment for cloud computing adoption, and to remove any vagueness surrounding this type of technology. Providers may need to clarify their position and stance when it comes to offering in-house services vs cloud services, which in turn can affect clients’ confidence. Prospectors appear to be willing to adopt cloud computing services despite security concerns, as they rely on the element of trust. Nevertheless, providers need to be aware of common concerns that SMEs experience when they make their adoption decision, such as locality, and trialability.

8 All these points can play a role in spreading a sense of trust and relief among SMEs to encourage adoption. In turn this will lead to the diffusion of cloud computing. Giving the SMEs the chance to try the products before the actual use would increase the awareness about cloud services. Also, an SME will be able to assess the services before the actual use regarding the compatibility with its existing systems and to evaluate the complexity level. Finally, since prospectors and most of the clients who are already adopting cloud services have stressed the importance of the physical location of the data centre, providers can take this point into consideration when choosing the place for their data centre, and also in their negotiations with the potential customers.

9 Future research could build on this study by examining cloud computing adoption in different sectors and industries and in different countries in both a qualitative and

quantitative way. More specifically, this study adopted a semi-structured interview data collection method to explore the decision-making process of SMEs. Although this approach is useful in delving into business-related decision analysis, it limits the ability to generalise. Also, future studies could consider interviewing different stakeholders (e.g. top managers, staff in the IT department, end users, etc.) within the same firm in the form of case studies. [10] Our sample was limited to one or two companies 266 JEIM 26,3 per sector. Having more participants per sector could better represent trends within each sector and allow for more reliable comparisons. This also applies to the geographical area considered. Different regions could be examined, both nationally and internationally. Given that SMEs are pervasive in all economies this will call for a careful selection of samples that can help provide a representative picture of cloud computing.

References

Further Reading

Appendix

About the authors

Yazn Alshamaila is a PhD student at the Newcastle University Business School. He received his MSc in Business Information Technology (BIT) from Northumbria University, UK. Currently, he is involved in Information and communication Technologies (ICT) innovation and his main research interest is investigating the adoption of cloud computing in small and medium-sized enterprises. Yazn Alshamaila is the corresponding author and can be contacted at: yazn.alshamaila@ncl.ac.uk

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Resuelva las siguientes consignas

Pre- lectura

1. Lea la fuente, ¿cuál es el tema del texto que va a leer?
2. Analice el paratexto, título, subtítulos, abstract. ¿Cómo está organizado el texto? ¿Qué tipo de información va a incluir cada sección?
3. De acuerdo al análisis anterior decida qué tipo de texto va a leer.

4. ¿Quiénes escriben este tipo de textos? ¿Con qué objetivos? ¿qué sección del texto puede darle información para contestar estas preguntas?
5. ¿Qué sección ofrece un resumen del contenido?
6. Analice la organización de esta sección.
7. ¿Qué subtítulo del abstract tendría que leer si quiero tener información sobre (hay un subtítulo extra):

a. Los alcances de este artículo de investigación	
b. Los objetivos	
c. Los resultados	
d. Las contribuciones	
e. Estudios previos	
f. El género del texto	
g. El enfoque teórico	
h. Conceptos relevantes	

Lectura

Concéntrese en la **Introducción**, lea las primeras oraciones de cada párrafo para conocer los temas que se desarrollarán en esta sección.

8. Haga un listado sobre la organización del artículo (tenga en cuenta el Párrafo 3)

9. **Conclusión:** lea los comienzos de cada párrafo y coloque una tilde (v) en cada tema que va a desarrollar (a-j). Hay tres temas que no corresponden a ningún párrafo:

<i>a. SUGERENCIAS PARA INVESTIGACIONES FUTURAS</i>	
<i>b. RECOLECCION DE DATOS</i>	
<i>c. POSIBLES IMPLICANCIAS DE LOS RESULTADOS</i>	
<i>d. PUBLICACIONES PREVIAS Y SU INFLUENCIA PARA ESTE ESTUDIO</i>	
<i>e. LIMITACIONES DEL ESTUDIO</i>	
<i>f. IMPACTO DEL ESTUDIO</i>	
<i>g. IMPORTANCIA DE LA TEMÀTICA</i>	
<i>h. VARIABLES QUE AFECTAN LA ADOPTACIÓN DE SERVICIOS EN LA NUBE</i>	
<i>i. OBJETIVO Y RESULTADO PRINCIPAL DEL ESTUDIO</i>	

j. CONTRASTE CON OTROS ESTUDIOS	
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10. ¿A qué se refiere “these findings”? (párrafo 7)

11. ¿Qué relación lógica establecen los conectores “On the other hand, however...” ¿Qué dos ideas están conectando?

Post-lectura

Escriba un párrafo con cuatro oraciones que sintetizen el contenido del artículo, previamente discuta con un compañero qué temas incluiría en esta síntesis.



Sustainable urban development: it's time cities give back

With dwindling natural resources to feed cities, urban planning needs to be regenerative from the start

Tuesday 13 August 2013

Our understanding of cities, in both theory and practice, is at a turning point. Urban areas across the world face complex and rapidly evolving challenges.

But what are the changes we need in order to transform our cities into liveable habitats? What shape does a city that inspires and engages citizens, governments and the public sector and private sectors to work together actually take? Is "sustainable urban development" still the best guiding concept for policy makers, urban planners, architects and investors in building the cities of the future?

The classic definition of sustainability comes from the Brundtland report of 1987: "Sustainable development is development that meets the needs of the present without compromising the ability of future generations to meet their own needs." But the term has since become like a rubber band that can be stretched in all directions. Whatever the interpretation, one thing is clear: Sustainable development is not enough anymore. This is because "the ability of future generations to meet their own needs" is already compromised.

Humanity is eating into the natural capital of the planet and eroding its resilience when it should be living off the income it regenerates. When we consider how degraded our ecosystems already are, there is far less to sustain today than there was 20 years ago. Not surprisingly, cities and urban areas are major contributors to this trend.

Over 400 million people in Africa currently live in urban areas, a number that will triple by 2050. Rapid urbanisation in many industrialising countries can be traced to push factors such as displacement due to violent conflict and a worsening of livelihoods due to natural disasters, including climate change. The mass migration into cities can also be attributed to the attractive pull factors of urban areas from improved access to food, healthcare and education and perceived prospects of a better life.

We can trace the increasing resource intensity of the urbanisation process back to the industrial revolution. Since then, disregard for resource generation and use has significantly contributed to climate change and the loss of soil carbon, natural fertility of farmland and biodiversity all over the world.

An overwhelming share of charcoal produced in rural areas in Africa is transported to urban areas to be used for cooking. This is one of countless examples illustrating the dependency of a metropolis on the area outside its borders. Almost everything we touch in a city is manufactured and originally sourced from the hinterland. That is, most of the raw materials a city absorbs are not generated from inside the city itself but rather from the rest of the planet. It is high time the city gave back.

Regenerative urban development

A new urban agenda is necessary in ensuring that cities not only become resource-efficient and low carbon-emitting, but go beyond that to positively enhance the ecosystems which provide them with goods and services. The solution lies in thinking beyond the vague and unambitious notion of sustainability and, instead, actively working towards regenerating soils, forests and watercourses. The aim is to improve rather than merely sustaining their currently degraded condition.

This new urban agenda transforms urban areas into regenerative cities that dramatically reduce their dependency on fossil fuels, boost the deployment of renewable energies, reintroduce water to the hydrological cycle and make sewage reprocessing and nutrient capture a central plank of urban waste management. A wide range of technical and management solutions towards this end are already available, but so far implementation has been too slow and too little.

From vision to reality

Enabling policy frameworks constitute the foundation of progressive social and economic development, achieving climate protection goals, and meeting ever-increasing global energy demand. The political environment plays a critical role in enabling businesses to invest in future-just and sustainable technology.

Transforming urban infrastructure into regenerative systems consequently requires an integrated approach, coordinated action and policy dialogue. It looks to straddle the public, private and civil society spheres in a multi-stakeholder participatory process. It also asks for a cross-sector approach among authorities.

Fiona Woo is policy officer of climate and energy at the World Future Council. Follow @Good Policies on Twitter. The 3rd annual Future of Cities Forum will provide a platform for multi-sector dialogue on the challenges and opportunities for regenerative urban development. For more information please visit www.futureofcitiesforum.com.

Lea el texto y responde las siguientes consignas en español:

- 1) ¿Cuál es la opinión del autor sobre el tercer interrogante que se plantea en la introducción y cómo la justifica?
- 2) Explique a qué se refiere el autor con los conceptos “push factors” y “pull factors”.
- 3) ¿Qué significa “It is high time the city gave back” en el contexto dado?
- 4) Describa en qué consiste la nueva “agenda” urbana a la que se refiere el texto.
- 5) ¿Qué se necesita para llevar a cabo dicha “agenda”?
- 6) Elabore una síntesis del texto que incluya el propósito del autor y los conceptos más importantes en un párrafo de tres oraciones que posea una redacción clara y coherente.

JANUARY 11, 2018

New 'emotional' robots aim to read human feelings



Buddy the companion robot by Blue Frog Robotics is seen on display during the CES Unveiled preview event at the Mandalay Bay Convention Center during CES 2018 in Las Vegas on Jan 7, 2018. Credit: AFP/Mandel Ngan

The robot called Forpheus does more than play a mean game of table tennis. It can read body language to gauge its opponent's ability and offer advice and encouragement. "It will try to understand your mood and your playing ability and predict a bit about your next shot," said Keith Kersten of Japan-based Omron Automation, which developed Forpheus to showcase its technology. "We don't sell ping pong robots but we are using Forpheus to show how technology works with people," said Kersten.

Forpheus is among several devices shown at this week's Consumer Electronics Show which highlight how robots can become more humanlike by acquiring "emotional intelligence" and empathy.

Although this specialization is still emerging, the notion of robotic empathy appeared to be a strong theme at the huge gathering of technology professionals in Las Vegas.

Honda, the Japanese auto giant, launched a new robotics program called Empower, Experience, Empathy including its new 3E-A18 robot which "shows compassion to humans with a variety of facial expressions," according to a statement.

Although empathy and emotional intelligence do not necessarily require a humanoid form, some robot makers have been working on form as well as function. "We're been working

very hard to have an emotional robot," said Jean-Michel Mourier of French-based Blue Frog Robotics, which makes the companion and social robot called Buddy, set to be released later this year. "He has a complex brain," Mourier said at a CES event. "It will ask for a caress or it will get mad if you poke him in the eye." Other robots such as Qihan Technology's Sanbot and SoftBank Robotics' Pepper, are being "humanized" by teaching them to read and react to people's emotional states.



The Omron Forpheus robot plays table tennis against a human at CES in Las Vegas. (AFP)

Pepper is "capable of interpreting a smile, a frown, your tone of voice, as well as the lexical field you use and non-verbal language such as the angle of your head," according to SoftBank.

Robot in human shoes

Developing emotional intelligence in robots is a difficult task, melding the use of computer "vision" to interpret objects and people and creating software that can respond accordingly.

"Empathy is the goal: the robot is putting itself in the shoes of the human, and that's about as hard as it gets," said Patrick Moorhead, a technology analyst with Moor Insights & Strategy.

"It's not just about technology, it's about psychology and trust." Moorhead said this technology is still in the early stages but holds promise in some areas, noting that there is strong interest in Japan amid a lack of caretakers for the elderly population. "In some ways it can be a bit creepy if you're crying and the robot is trying to console you," he said. "If you have no friends, the next best thing is a friend robot, and introverts might feel more comfortable talking to a robot."

'Emotion chip'

One CES exhibitor offered a promise of going further than the current devices by developing an "emotion chip" which can allow robots to process emotions in a manner similar to humans.

"There has been a lot of research on detecting human emotions. We do the opposite. We synthesize emotions for the machine," said Patrick Levy-Rosenthal, founder of New York-based Emoshape, which is producing its chip for partners in gaming, virtual and augmented reality and other sectors. It could be used to power a humanoid robot, or other devices. For example, an e-reader could better understand a text to infuse more emotion in storytelling.

As for Forpheus, Kersten said the robot's ability to help people improve their table tennis skills could have numerous applications for sports, businesses and more. "You could sense how people are feeling, if they are attentive or in a good state to drive," he said.

Another key application could be in health care, he said: "In an elderly patient facility, you can determine if someone is in distress and needs help."

<https://techxplore.com/>

Lea el artículo y responde las siguientes consignas:

1. Explique las características de los robots mencionados.
2. ¿Qué quiere decir el autor con el subtítulo "Robots in human shoes"?
3. En el cuarto párrafo se encuentra el conector **ALTHOUGH** marcado en negrita. ¿Qué tipo de conector es? ¿Cuáles son los dos conceptos que relaciona?

4. Sintetice el artículo en no más de cuatro oraciones, teniendo en cuenta los conceptos principales.